

# README — data\_vault/

## The Living Semantic Knowledge Base, High-Speed Cache & Cognitive Tiers

### W5+H Subsystem Identity

- **WHO:** Governed directly by the Files Executive Agent via the OpenWiki TUI.
  - **WHAT:** Houses the 5+1 Cognitive Memory Tiers: immutable cold sources, conceptual wiki markdown, machine recall caches, procedural tool libraries, and episodic telemetry.
  - **WHEN:** Queried during every reasoning pass; updated dynamically upon new document ingestion or task execution.
  - **WHERE:** [novae-xorpus/data\\_vault/](#) (Federated Subsystem Root).
  - **WHY:** Enforces strict separation between human-readable markdown and machine-readable vector/JSONL caches, preventing context pollution and database degradation.
  - **HOW:** Synchronizes nodes via bidirectional [\[\[wikilinks\]\]](#), computes SHA256 checksums, and maintains localized tier manifests.
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### Internal Directory Topology

data\_vault/

|                   |                                                         |
|-------------------|---------------------------------------------------------|
| — README.md       | # This document (Cognitive memory tiers & access rules) |
| — manifest.jsonl  | # Master catalog of semantic notes and cache shards     |
|                   |                                                         |
| — 01_raw_sources/ | # TIER 1: Immutable Ground Truth (Cold Archive)         |
| — manifest.jsonl  | # Local raw asset index                                 |
| — pdf/            | # Source whitepapers, specs, data sheets (*.pdf)        |
| — media/          | # Audio memos, screen vision captures (*.wav, *.mp4)    |

|                       |                                                             |
|-----------------------|-------------------------------------------------------------|
| └─ text/              | # Raw unformatted scrapes & exports (*.txt, *.html)         |
|                       |                                                             |
| └─ 02_wiki_md/        | # TIER 2: Semantic Memory (The LLM Wiki)                    |
| └─ manifest.jsonl     | # Local conceptual node index                               |
| └─ concepts/          | # Atomic linked notes (Zettelkasten domain models)          |
| └─ architectures/     | # Technical specs & cross-repo blueprints                   |
| └─ entities/          | # Device registers, schema contracts, hardware profiles     |
| └─ indexes/           | # Maps of Content (MOCs) & taxonomy hubs                    |
|                       |                                                             |
| └─ 03_recall_cache/   | # TIER 3: Working Memory Accelerator (High-Speed Cache)     |
| └─ manifest.jsonl     | # Local cache shard index                                   |
| └─ jsonl/             | # Pre-tokenized chunks for low-latency prompt injection     |
| └─ vectors/           | # Dense embedding indices (FAISS / HNSW checkpoints)        |
| └─ kv_store/          | # Embedded SQLite key-value fast lookup tables              |
|                       |                                                             |
| └─ 04_skills_runtime/ | # TIER 4: Procedural Memory (Skills & Executables)          |
| └─ manifest.jsonl     | # Local procedural catalog                                  |
| └─ prompt_skills/     | # Modular SKILL.md prompt definitions (*.md, *.yaml)        |
| └─ extracted_tools/   | # Mined deterministic code blocks (cli/*.sh, wrappers/*.py) |
| └─ runtimes/          | # Operational daemon scripts & socket listeners             |
| └─ policies/          | # Self-correction heuristics & validation schemas           |

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|-----------------------|--------------------------------------------------------|
| └─ 05_episodic_logs/  | # TIER 5: Episodic Memory (Telemetry & Telemetry Logs) |
| ├─ manifest.jsonl     | # Local session & audit registry                       |
| ├─ daily_driver_sync/ | # Ingested logs from mobile edge sessions              |
| ├─ trajectories/      | # Tri-model run traces (Query, Executor, Frontier)     |
| ├─ rlvr_verifiers/    | # Graded rewards (+1.0 / -1.0) & assertion outcomes    |
| └─ hygiene_reports/   | # Schema integrity audits & failure logs               |

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## Beginner-Proof Implementation Rules

1. **Sensory Immutability:** Files placed in `01_raw_sources/` are write-once. Never overwrite or reformat originals.
2. **Clean Separation:** Never place raw tabular JSONL or binary embeddings into `02_wiki_md/`. Machine caches belong strictly in `03_recall_cache/`.